A VISUAL INVESTIGATION

Brain to AI

How did humans begin building artificial intelligence?

By looking inward — at the way our own biological intelligence already works.

Nine pages, fifteen figures, and one boundary. Every figure here is drawn to be studied on its own, and every figure states what it is simplifying.

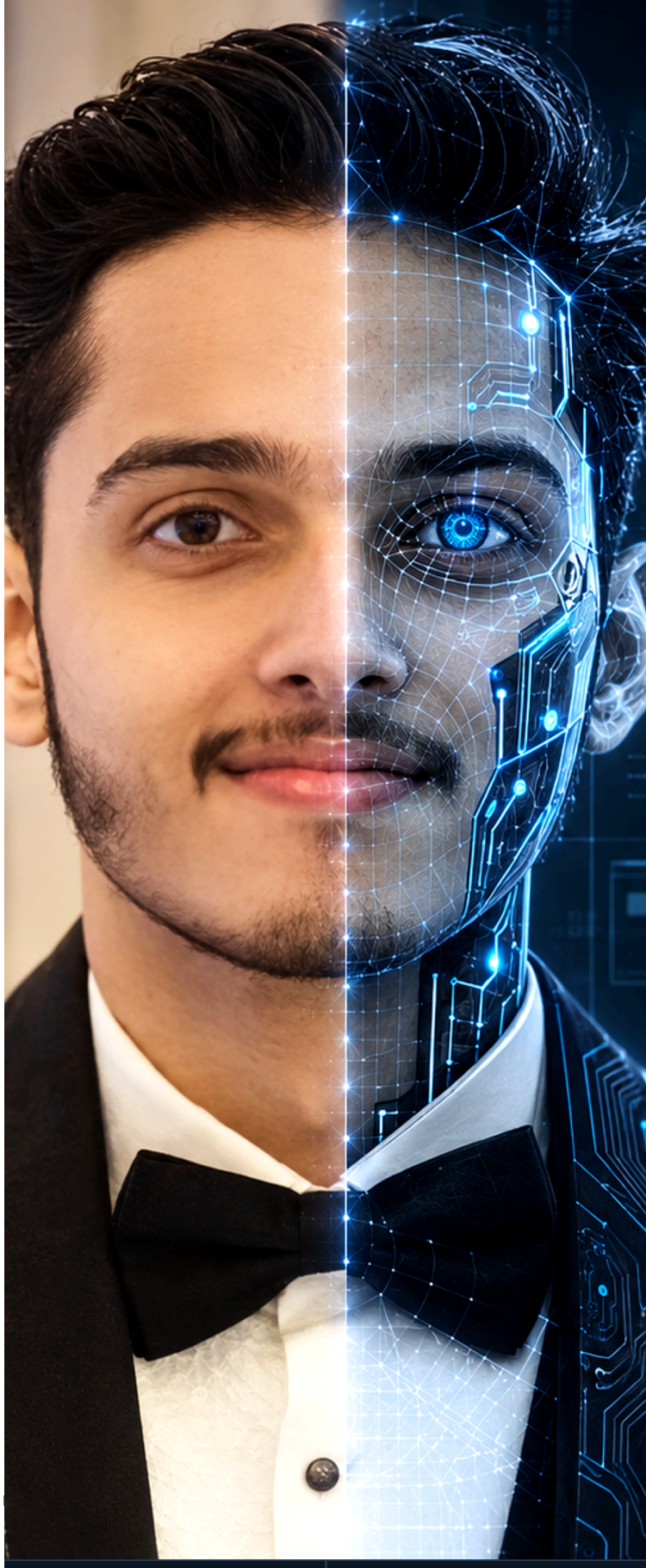


FIG. 1 — WHAT THIS ARTICLE TRACES BIOLOGY → ABSTRACTION → MAC

01 **Biological system**
Brain, neuron, synapse, network, plasticity · p. 02–04

02 **Selective abstraction**
Five principles carried across; the rest set aside · p. 05

03 **Engineered system**
Artificial neuron, learning, depth, modern AI · p. 06–08

04 **The distance between them**
What holds, what the evidence says, the verdict · p. 09

OBSERVED
Biological intelligence
Evolved, embodied, still only partly understood

ENGINEERED
Artificial intelligence
Specified, trained, differentiable end to end

The split is a relationship, not an identity.

Artificial neural networks are inspired by selected abstractions of biological systems. They are not biological replicas — and the shared vocabulary, neuron and learning and attention, is a naming convention rather than a claim of shared mechanism.

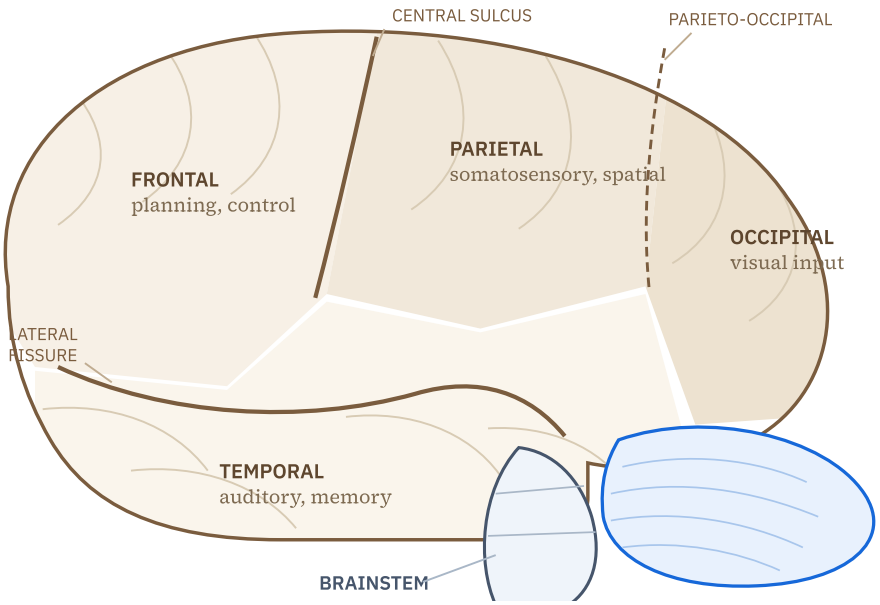
Part I begins overleaf

The organ, and why it is the starting point

Open a brain and you find no control room. You find tissue organised into regions whose activity correlates with particular functions — and which never work alone. Every region is built from the same class of component, wired differently, which is why the investigation cannot stop at anatomy.

FIG. 2 — PANEL A · LATERAL SCHEMATIC OF THE LEFT HEMISPHERE

ANTERIOR LEFT · NOT TO SCALE

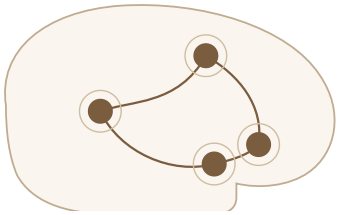


CORTEX
2–4 mm of folded tissue; folding fits area into volume.

SULCUS = GROOVE
GYRUS = RIDGE

LOBE BOUNDARIES
follow landmarks, not function.

PANEL B · ONE TASK, MANY REGIONS



READING A WORD

visual cortex · form

temporal · sound, meaning

frontal · control, speech

plus subcortical and cerebellar contributions

NO REGION IS AN "INTELLIGENCE CENTRE"

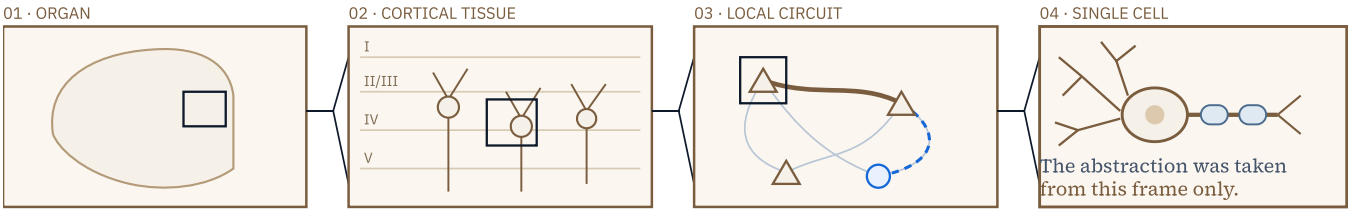
ORDER-OF-MAGNITUDE FACTS

~10^{11}	neurons. Published counts range 62–94 billion and do not agree.
~10^{14}	synapses. Never counted; estimated from density.
~20 W	to run all of it, resting or working hard.

I have quoted "86 billion neurons" myself. Checking it was a lesson in reading sources: it comes from four brains, its own interval spans 73–99 billion, and a second study using the same method lands at 61–73. An estimate wearing the clothes of a measurement.

FIG. 3 — THE CONTROLLED ZOOM · ORGAN → TISSUE → CIRCUIT → CELL

FOUR MAGNIFICATIONS



≈ 15 cm · ≈ 2–4 mm · SIX LAYERS · ▲ EXC ● INH · ≈ 100–500 µm · ≈ 10–100 µm

WHAT THIS SHOWS Structure at one scale does not explain behaviour at another. Frames 02 and 03 are where most of the open questions still live — and they are precisely the levels the artificial version never modelled.

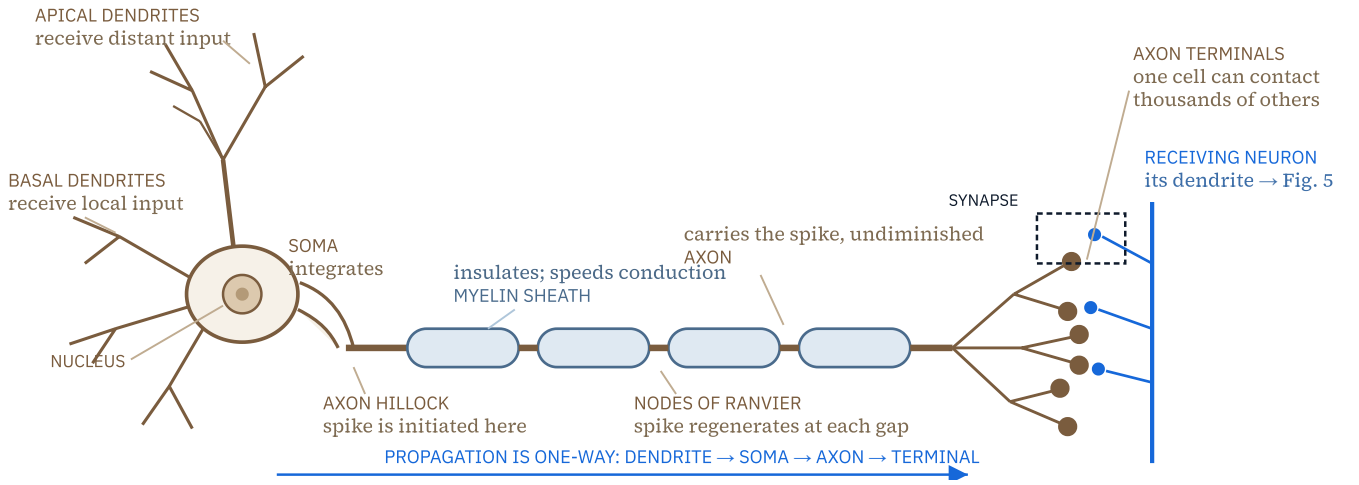
SIMPLIFIED Schematic, not micrographs. Cell densities and connection counts are reduced by orders of magnitude. Glia, neuromodulators and blood supply appear nowhere.

The neuron, and the junction between two of them

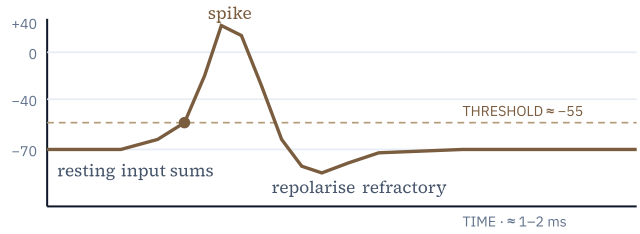
A living cell specialised for signalling: it collects input across a branched dendritic tree, integrates it at the cell body, and — if the membrane at the axon hillock crosses threshold — fires an all-or-none spike down the axon. Fig. 5 enlarges the junction where that spike meets the next cell: the structure the artificial model replaces with one number.

FIG. 4 — ANATOMY AND DIRECTION OF PROPAGATION

CORTICAL PYRAMIDAL CELL, SCHEMATIC



MEMBRANE POTENTIAL DURING ONE SPIKE



All-or-none: below threshold, nothing leaves.

WHY THIS IS NOT AN ARITHMETIC UNIT

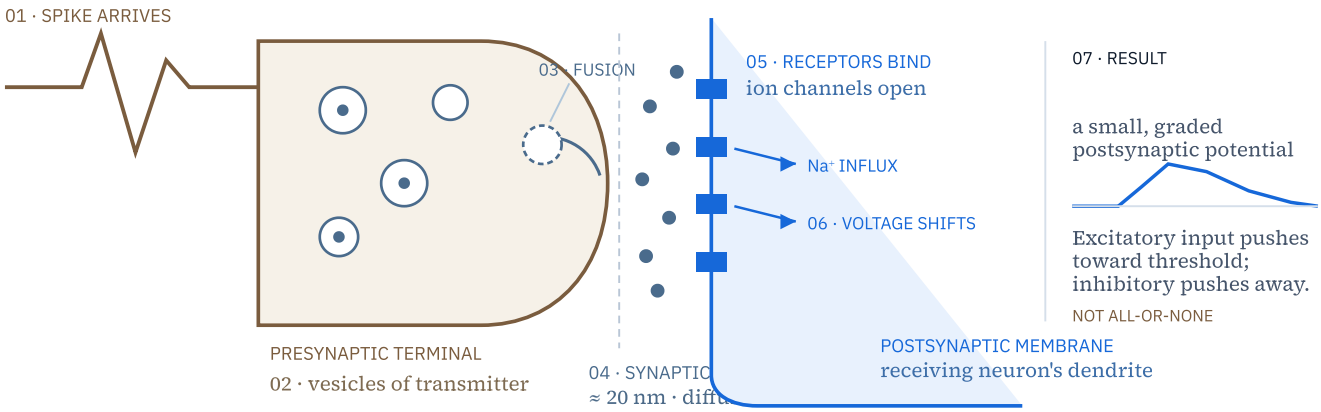
Dendrites are not passive wires: input location, timing and local nonlinear conductances all change what reaches the soma, so integration is not a plain weighted sum.

A measured consequence: reproducing the input/output behaviour of *one* simulated layer-5 pyramidal cell at millisecond resolution required a deep network of **5–8 layers**. Remove its NMDA receptors and one hidden layer suffices.

BENIAGUEV, SEGEV & LONDON · NEURON 2021

FIG. 5 — CROSS-SECTION OF A CHEMICAL SYNAPSE

ENLARGED FROM FIG. 4



WHAT THIS SHOWS Synaptic transmission is a biological electrochemical process with structure at every step: probabilistic release, diffusion delay, receptor kinetics, reuptake. "Connection strength" is an emergent summary of all of it — and EM reconstruction puts that strength at roughly **26 distinguishable steps ≈ 4.7 bits**, not one number.

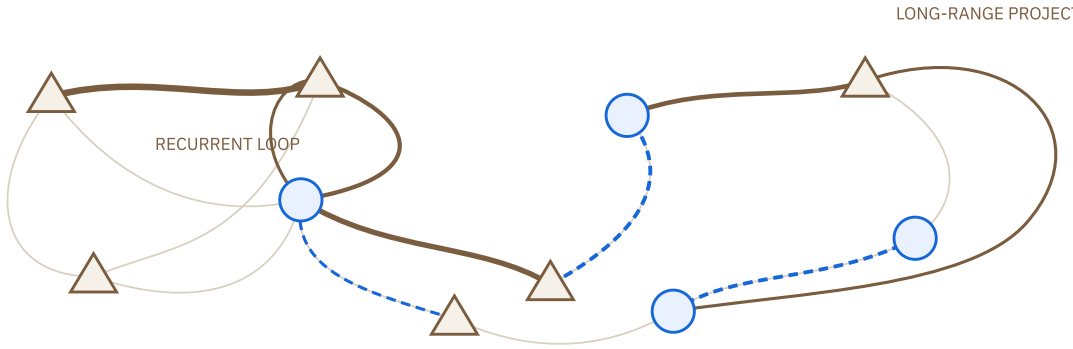
OMITTED Electrical synapses, transmitter variety, astrocytic regulation, receptor subtypes with different time courses, and short-term facilitation and depression. Bits figure: rat hippocampus, Bartol et al. 2015.

The network, and how experience changes it

Cortical wiring is recurrent and densely reciprocal, with excitatory and inhibitory cells intermixed — compare Fig. 6 with the tidy artificial network on page 06. Fig. 7 shows the mechanism that carried across: co-activation changes the efficacy of an existing junction.

FIG. 6 — RECURRENT, MIXED EXCITATORY / INHIBITORY CIRCUIT

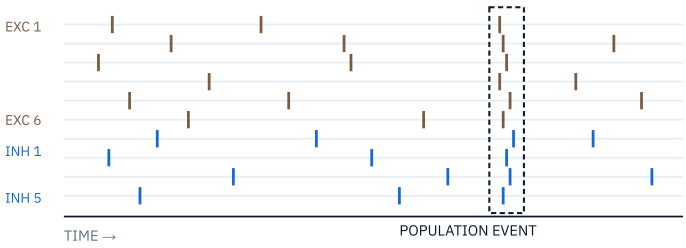
LINE WEIGHT = CONNECTION STRENGTH



▲ EXCITATORY (PYRAMIDAL) · RAISES TARGET TOWARD THRESHOLD
● INHIBITORY (INTERNEURON) · DASHED LINES · PUSHES AWAY FROM IT

DALE'S LAW
A cell is excitatory or inhibitory, not both.

DISTRIBUTED ACTIVITY · SPIKE RASTER, 11 CELLS



WHAT THIS SHOWS Activity is sparse and distributed; timing itself carries information — exactly what the abstraction discards when it replaces spikes with a continuous rate.

FROM CELLS TO NETWORK BEHAVIOUR

A single neuron does almost nothing interesting. Recognising a face or catching a ball lives in the pattern of connections between very large numbers of simple units. Intelligence of this kind is a property of the network, not of the part — the one structural principle the artificial version inherited more or less intact.

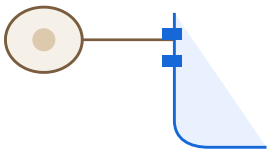
LIMITS OF FIG. 6

Ten cells stand in for millions; a real cortical neuron makes thousands of synapses, not four. The layout is spatial convenience, not anatomy. And connectivity alone does not explain function — four decades after the 302-neuron *C. elegans* diagram, its behaviour is still not fully predicted from it.

FIG. 7 — SYNAPTIC PLASTICITY IN FOUR PANELS

CONCEPTUAL · ONE MECHANISM AMONG SEVERAL

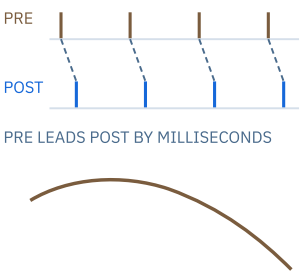
A · INITIAL STATE



Few receptors, small contact area. The input barely moves the cell.

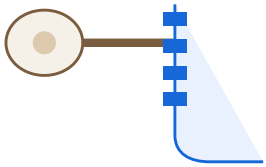
WEAK EFFICACY

B · REPEATED CO-ACTIVATION



Order and interval matter. Pre-before-post strengthens; reverse it and it weakens.

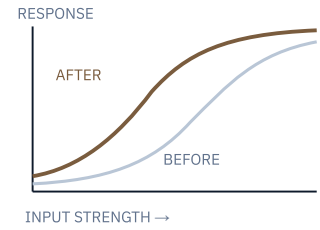
C · AFTER · CHANGED EFFICACY



More receptors, larger contact. Same spike, larger response.

WIRING UNCHANGED

D · INTERPRETATION



Experience writes no record. It shifts how strongly one cell influences the next — changing every future response of the network.

WHAT THIS SHOWS Learning is a physical change at existing junctions, driven by correlated activity and sensitive to timing. Panels A and C are the same circuit. Hebb, 1949 — and the load-bearing idea for everything in Part II.

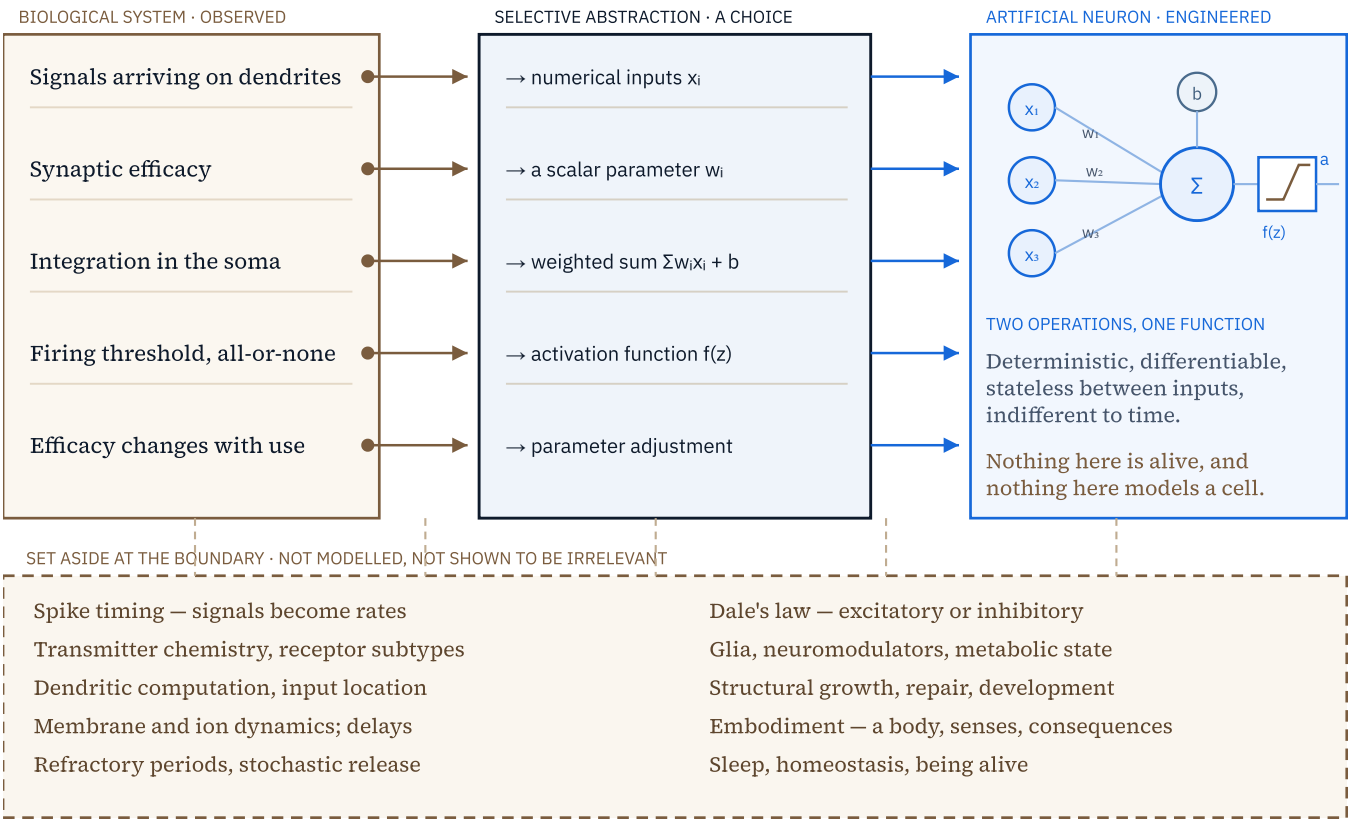
NOT SHOWN Homeostatic scaling, structural rewiring, neuromodulatory gating, systems consolidation during sleep — several distinct mechanisms, none reducible to this panel. Note also that this rule is *local*; the algorithm on page 07 is not.

What was carried across, and what was left behind

In 1943 McCulloch and Pitts asked whether a neuron could be described mathematically. In 1958 Rosenblatt built a machine on that description. Neither was trying to build a brain; both were testing whether one principle could be lifted out of biology. This figure is the accounting of that lift — five properties carried over, the rest set aside deliberately.

FIG. 8 — THE SELECTIVE ABSTRACTION · BIOLOGICAL PROPERTY → MATHEMATICAL OPERATION

A MAPPING, NOT AN EQUIVALENCE



WHAT THIS SHOWS Artificial neural networks are inspired by five selected principles. The middle column is a decision, made by named people who said so at the time — not a discovery that biology reduces to arithmetic. The dashed box is the price of that decision, and it has never been paid back.

READ THE ARROWS as "was replaced by", never as "is the same as".

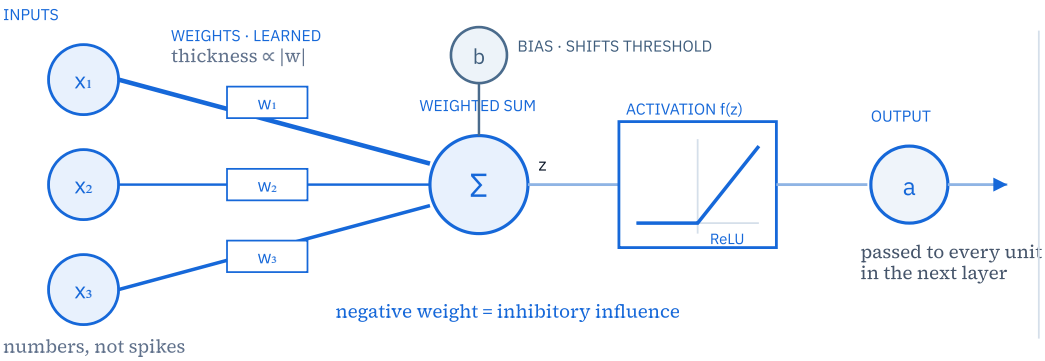
The right-hand column is not a description of a neuron. It is a decision about which one of its properties to keep — reasonable, deliberate, and misleading only when later readers mistake the abstraction for the thing.

Everything that follows is built from the middle column. It works extraordinarily well. Keeping the dashed box in view is what stops "inspired by the brain" from sliding into "works like the brain".

The artificial neuron, and the same unit in layers

Three inputs, three weights, one bias, one non-linearity, one output. Stack the unit and information is rewritten layer by layer: early layers pick up simple structure, later ones combine it into something abstract. Nobody programs those stages — they are what the weights settle into.

FIG. 9 — ONE UNIT, EVERY TERM LABELLED



THE WHOLE UNIT

$z = \Sigma w_i x_i + b$

$a = f(z)$

Two lines. Everything that follows is this, repeated and stacked.

w AND b ARE THE ONLY THINGS THAT LEARN

ONE WORKED EXAMPLE

inputs	x = [1.0, 0.5, 2.0]
weights	w = [0.4, −0.3, 0.2]
bias	b = 0.1

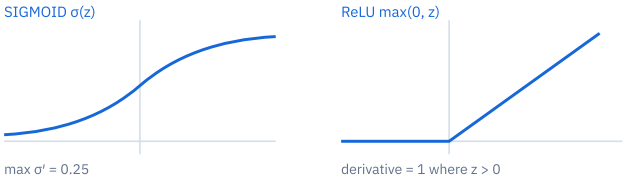
z = 0.4(1.0) − 0.3(0.5) + 0.2(2.0) + 0.1

z = 0.4 − 0.15 + 0.4 + 0.1 = **0.75**

a = ReLU(0.75) = **0.75**

Had z come out negative, ReLU would return 0 and this unit would contribute nothing at all to the next layer — for this input.

WHY THE NON-LINEARITY IS NOT DECORATION

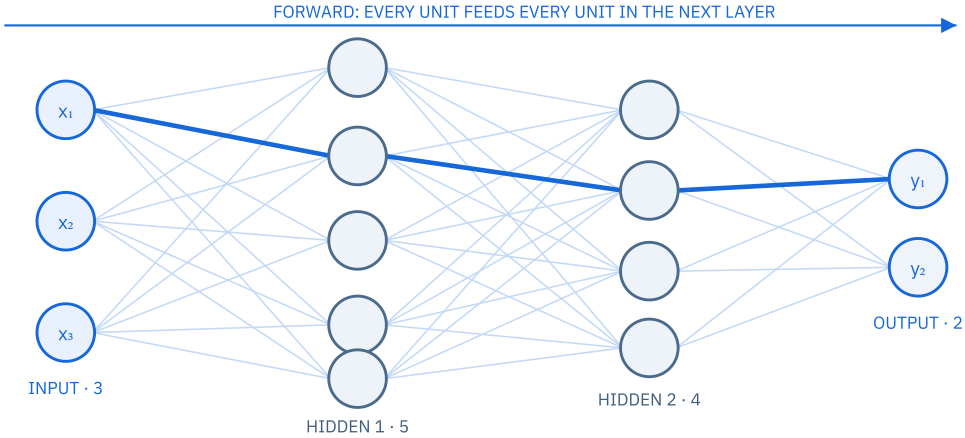


Stack two purely linear layers and you get $W_2(W_1x) = (W_2W_1)x$ — one linear layer in a costume. Without a non-linearity, depth buys nothing.

And the choice between these curves decided a decade. Sigmoid's derivative peaks at 0.25, so gradients shrink at least fourfold per layer going back; after ten layers the earliest weights receive about a millionth of the signal. ReLU's is exactly 1 in the positive region — the shrinking stops and depth becomes trainable.

FIG. 10 — FULLY CONNECTED FEEDFORWARD NETWORK · 3 → 5 → 4 → 2

COMPARE WITH FIG. 6



PARAMETER COUNT

3×5 + 5 = 20

5×4 + 4 = 24

4×2 + 2 = 10

54 parameters

A frontier model has on the order of 10^{11}.

SAME UNIT. MORE OF IT.

WHAT THIS SHOWS Information enters at one end and is rewritten layer by layer. The values in a hidden layer are not labelled here because nobody assigned them meanings — they are whatever reduced the loss.

HOW IT DIFFERS FROM FIG. 6 No recurrence, no inhibitory cell type, no timing, and perfectly regular connectivity no tissue exhibits. The tidiness is the abstraction showing.

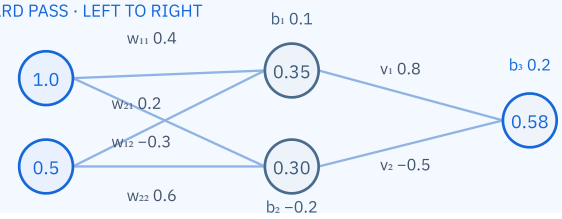
The forward pass, the error, and the update

Activations travel forward; gradients travel back along the same weights. Separating the two channels is the only way to see what backpropagation actually does — and the three properties of it that no cortex is known to have. Fig. 12 then follows a single parameter through one step.

FIG. 11 — FORWARD PROPAGATION AND BACKPROPAGATION ON THE SAME NETWORK

2 → 2 → 1 · ALL VALUES COMPUTED

FORWARD PASS · LEFT TO RIGHT



$z_1 = 0.4(1.0) - 0.3(0.5) + 0.1 = 0.35 \rightarrow a_1 = 0.35$

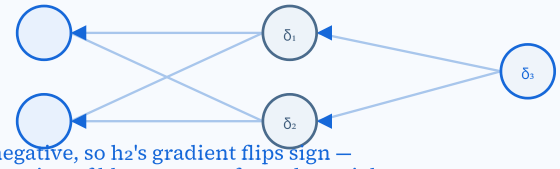
$z_2 = 0.2(1.0) + 0.6(0.5) - 0.2 = 0.30 \rightarrow a_2 = 0.30$

$z_3 = 0.8(0.35) - 0.5(0.30) + 0.2 = 0.33$

$\hat{y} = \sigma(0.33) = 0.5818$ · true label $y = 1$

$L = -\log(0.5818) = 0.5417$

BACKWARD PASS · RIGHT TO LEFT, SAME WEIGHTS



v_2 is negative, so h_2's gradient flips sign — the direction of blame comes from the weight.

$\partial L/\partial z_3 = \hat{y} - y = -0.4182$ (σ' cancels)

$\partial L/\partial v_1 = -0.4182 \times 0.35 = -0.1464$

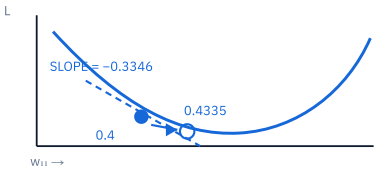
$\delta_1 = -0.4182 \times 0.8 = -0.3346$ · $\delta_2 = +0.2091$

$\partial L/\partial w_{11} = -0.3346$ $\partial L/\partial w_{12} = -0.1673$

$\partial L/\partial w_{21} = +0.2091$ $\partial L/\partial w_{22} = +0.1046$

ONE NUMBER PER PARAMETER, EVERY STEP

FIG. 12 — ONE PARAMETER UPDATE, IN CLOSE-UP



CURRENT	0.4000
GRADIENT	−0.3346
RATE	η = 0.1
UPDATED	**0.4335**

$\mathbf{w \leftarrow w - \eta\, \partial L/\partial w}$

WHAT THIS SHOWS Learning is iterative adjustment, not insight. Negative slope, so the weight increases; run the example again and the loss falls from 0.5417 to 0.4876. Simplified: the real loss surface has as many dimensions as the model has parameters, and is not this smooth.

WHY NO CORTEX DOES THIS

1 · Weight transport. The backward pass reuses the exact forward weights; a biological feedback pathway would have to know them.

2 · Signed error. Gradients are signed, precise, sometimes extreme. Spikes are none of those.

3 · Silent feedback. In backprop the backward pass changes no activity; in cortex feedback changes activity by definition.

Plausible substitutes exist — feedback alignment, target propagation, equilibrium propagation, predictive coding — but none has replaced backprop, whose *principles* may still describe cortical learning even where the literal algorithm cannot.

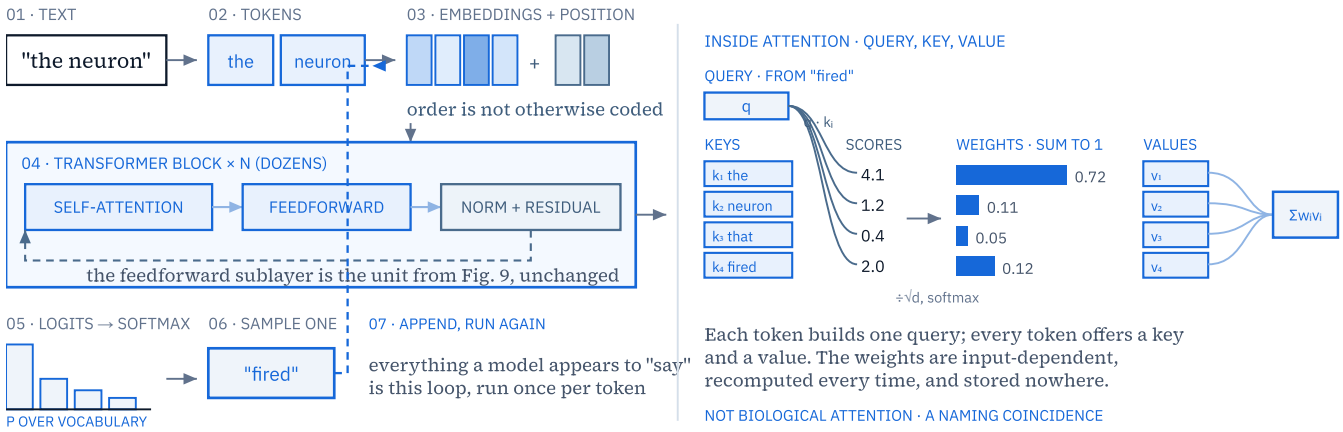
LILLICRAP, SANTORO, MARRIS, AKERMAN & HINTON · 2020

The unit never changed. The wiring did.

Convolutional networks exploit the fact that nearby pixels are related. Recurrent networks carry state along a sequence. The transformer, introduced in 2017, replaced that sequential carrying with attention — every element compares itself against every other, in parallel. Nothing about that mechanism was taken from neuroscience.

FIG. 13 — LANGUAGE-MODEL PIPELINE

DECODER-ONLY · ATTENTION INSET AT RIGHT



WHAT THIS SHOWS A pipeline of matrix operations and one learned probability distribution. The unit never changed — only the wiring and the scale. Remarkable engineering, not a description of thinking.

WHERE THE LINEAGE ENDS Attention was derived from a sequence-modelling problem, not from neuroscience. Nothing on this page has a biological antecedent.

WHAT SCALED

Parameters, data and compute. The unit — weights, bias, non-linearity — did not change at all, and neither did the learning rule from page 07.

WHERE THE LINEAGE ENDS

Attention answers a sequence-modelling problem: it removed the bottleneck that made long-range dependencies expensive. It is not a model of human attention, and the shared word is a naming coincidence.

A pipeline of matrix operations and one learned probability distribution. Remarkable engineering — and not a description of thinking.

Inspired by, not made of

Each row of Fig. 14 is a resemblance someone chose to exploit: what genuinely holds, then what the measurements say. Fig. 15 sets the two systems side by side on eight axes. Neither is a scoreboard.

FIG. 14 — THE ANALOGY, AUDITED CLAIM BY CLAIM

OBSERVATION · MECHANISM · EVIDENCE

CLAIM	WHAT HOLDS	WHAT THE EVIDENCE SAYS
"A neuron is a simple unit"	Receive, integrate, threshold, transmit sketches the gross behaviour.	A **5–8 layer** network was needed to reproduce one simulated layer-5 pyramidal cell at millisecond resolution; without NMDA receptors, one layer suffices. BENIAGUEV 2021
"A synapse is a weight"	Strength is real, adjustable, and is where learning lives in both.	Paired spines differ by only ~8% in size, implying about **26 distinguishable strengths ≈ 4.7 bits** — an order of magnitude more precision than assumed. BARTOL 2015
"It learns the way you do"	Both change connection strengths from experience; feedback fills the cortex.	The sharpest break — see Fig. 11. **No direct evidence** the brain runs a backprop-like algorithm, and the algorithm needs weight symmetry, signed errors and silent feedback. LILLICRAP 2020
"Same signals"	Both propagate activity through richly connected populations.	Real neurons signal in timed spikes, not rates; dendrites compute rather than collect; a cortical cell is excitatory *or* inhibitory, where an artificial unit is freely both.
"Comparable learning"	Both build reusable internal features nobody programmed.	Humans learn from few examples, carry a world model nobody taught them, and can be held responsible. The brain also runs on about **20 watts** where training runs on megawatts.

THE COUNTERWEIGHT Backprop-trained networks predict inferotemporal and auditory cortex responses better than models built on hand-designed biological priors, and the 2025 MICrONS data found a "like-to-like" connectivity rule that also emerges in artificial networks.

AND THE MAP WE LACK The abstraction was made in 1943. The first complete wiring diagram of a brain belonging to an animal that walks and sees arrived in 2024 — a fruit fly. The human brain has never been read end to end.

FIG. 15 — BIOLOGICAL AND ARTIFICIAL INTELLIGENCE, EIGHT AXES

DIFFERENT KINDS OF SYSTEM, NOT TWO SCORES

AXIS	BIOLOGICAL	ARTIFICIAL
Substrate	Living tissue; electrochemical, self-maintaining	Silicon; numerical operations on stored parameters
Basic unit	A cell with dendritic computation; thousands of types	A weighted sum and one non-linearity; one type
Connectivity	Recurrent, mixed E/I, local and long-range	Mostly feedforward; regular, architect-specified
Signals	Timed spikes; timing itself carries information	Continuous values; no intrinsic notion of time
Learning rule	Local, correlational, several mechanisms; not fully known	Global gradient descent on a specified loss
Embodiment	A body, senses, consequences; learning is situated	None. Inputs are datasets, outputs are predictions
Energy	~20 W total, sparse activity	Megawatts at training scale; dense activity
Development	Evolution plus a lifetime; plastic throughout, learns from few examples	One training phase on huge corpora, then frozen unless retrained
Responsibility	Can be asked why, and can be held to the answer	Produces an output; a person remains answerable for it

WHAT THIS SHOWS No row is a deficiency. Similar terminology does not imply identical mechanisms; a system can reproduce useful patterns without experience, understanding or responsibility.

We built artificial intelligence by taking inspiration from biological intelligence. Understanding the difference between the two may be just as important as building the next generation of AI.

ONE BRIDGE, AND THEN I STOP

If AI is a human-built abstraction inspired by a system we still do not fully understand, confidence in its behaviour cannot rest on assuming the analogy is complete. It has to be established another way — the subject of the next article, *The Untestable Machine*.

WHAT I WOULD TELL MYSELF

Go one level below the level you were given.

Check the number everybody repeats — it is the one nobody has checked.

When two fields share a word, assume they mean different things by it.

SOURCES Azevedo et al., *J Comp Neurol* 513:532 (2009) · Goriely, *Brain* 148:689 (2024) · Hebb (1949) · McCulloch & Pitts (1943) · Rosenblatt (1958) · Rumelhart, Hinton & Williams, *Nature* 323:533 (1986) · Bartol et al., *eLife* 4:e10778 (2015) · Samavat et al., *Neural Computation* 36:781 (2024) · Beniaguev, Segev & London, *Neuron* 109:2727 (2021) · Lillicrap, Santoro, Marris, Akerman & Hinton, *Nat Rev Neurosci* 21:335 (2020) · Vaswani et al. (2017) · Dorkenwald et al. and Schlegel et al., *Nature* 634 (Oct 2024) · MICrONS Consortium, *Nature* 640:435 and Ding et al., *Nature* 640:459 (Apr 2025) · *Nature Methods* 22, Method of the Year 2025.